

Math 2650 Recitation – Week 1 Concept Practice

Choose the section matching the skill you want to practice. Answers are on the back.

A. Coordinate Systems and Surface Equations

- A1.** A sphere has diameter endpoints

$$D_1 = (-1, 2, 0), \quad D_2 = (3, 2, 4).$$

Find an equation of the sphere in rectangular coordinates. Then rewrite the same sphere in cylindrical coordinates.

- A2.** A surface is given in spherical coordinates by

$$\rho \sin \phi (2 \cos \theta - \sin \theta) = 6.$$

Rewrite the surface in rectangular coordinates and identify the type of surface.

- A3.** Sphere S_1 is given by

$$(x - 1)^2 + (y + 2)^2 + (z - 2)^2 = 9.$$

Sphere S_2 has twice the radius of S_1 . The center of S_2 is one unit from the center of S_1 in the direction of the unit vector

$$\mathbf{u} = \frac{1}{3} \langle 2, -1, 2 \rangle.$$

Find a rectangular-coordinate equation for S_2 .

- A4.** The surface

$$(x - 1)^2 + (z + 2)^2 = 9$$

is a circular cylinder. State which coordinate axis its axis is parallel to, give one point on its axis, and rewrite the equation in cylindrical coordinates.

B. Vectors and Lines in Space

- B1.** Let

$$\mathbf{u} = \langle 1, 1, 0 \rangle, \quad \mathbf{v} = \langle 0, 1, 1 \rangle, \quad \mathbf{w} = \langle 1, -1, 1 \rangle.$$

Find (a, b, c) such that

$$\langle 7, 0, 5 \rangle = a\mathbf{u} + b\mathbf{v} + c\mathbf{w}.$$

- B2.** A line through (x_0, y_0, z_0) with direction vector $\langle a, b, c \rangle$ has parametric equations

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

Let ℓ be the line through $A = (1, 0, -1)$ and $B = (5, 2, 3)$, and let $C = (4, 1, 0)$. Find the point $P \in \ell$ such that $\overrightarrow{CP}$ is perpendicular to ℓ .

C. Sphere/Cone Geometry and Volume

- C1.** Sphere S_1 has center $(1, -2, 3)$ and radius 2, while sphere S_2 has center $(7, 2, 6)$ and radius 1. Find the *greatest possible distance* between a point on S_1 and a point on S_2 .

- C2.** Three concentric spheres S_1, S_2, S_3 have radii 6, R , and 2, respectively, with S_1 the largest. Choose R so that the volume inside S_1 but outside S_2 is equal to the volume inside S_2 but outside S_3 .

- C3.** Find the volume of the solid described in cylindrical coordinates by

$$-\sqrt{9 - r^2} \leq z \leq 4 - \frac{4}{3}r.$$

Answer Key

Final answers only

A. Coordinate Systems and Surface Equations

A1. Rectangular:

$$(x-1)^2 + (y-2)^2 + (z-2)^2 = 8.$$

Cylindrical:

$$(r \cos \theta - 1)^2 + (r \sin \theta - 2)^2 + (z - 2)^2 = 8.$$

A2.

$$2x - y = 6; \quad \text{plane}.$$

A3.

$$\left(x - \frac{5}{3}\right)^2 + \left(y + \frac{7}{3}\right)^2 + \left(z - \frac{8}{3}\right)^2 = 36.$$

A4. Axis parallel to the

$$y\text{-axis}.$$

One point on the axis is

$$(1, 0, -2).$$

Cylindrical:

$$(r \cos \theta - 1)^2 + (z + 2)^2 = 9.$$

B. Vectors and Lines in Space

B1.

$$(a, b, c) = (3, 1, 4).$$

B2.

$$P = (3, 1, 1).$$

C. Sphere Geometry and Volume

C1.

$$\sqrt{61} + 3.$$

C2.

$$R = 2\sqrt[3]{14}.$$

C3.

$$30\pi.$$